

CHRIS BHAILA

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OBJECTIVE

AI/ML engineer with hands-on experience building ML systems from scratch (unsupervised clustering, decision trees, CNNs, semantic search) and shipping production full-stack applications, seeking an internship/role in agentic AI development — building LLM-powered systems capable of autonomous reasoning, tool use, and task execution.

EDUCATION

Bachelor's in Computer Application 2021 - 2026

Thames International College, Kathmandu

Relevant Coursework: Web Technology, MySQL, Scripting Language, Computer Graphics, Data Structures and Algorithms, Probability and Statistics, Digital Logic, Numerical Methods, DotNet Technology, OOP in Java, Financial Accounting, Software Project Management, Artificial Intelligence

+2 2019 - 2021

Management with Computer Science

Little Angels' College of Management, Hattiban, Lalitpur, GPA: 3.38

Relevant Coursework: Economics, Calculus, C Programming

SKILLS

AI/ML	PyTorch, scikit-learn, Sentence-BERT/Transformers, NumPy, Pandas, Keras, Prompt Engineering
Programming	Python, JavaScript, PHP, Java, C, C#
Web/Backend	Django, Laravel, REST APIs, PostgreSQL, MySQL
Tools	Git, Jupyter, MATLAB
Languages	English, Nepali, Newari

INTERNSHIP

Full-Stack Developer Intern 2026
Nahusha Investment Pvt. Ltd. Kathmandu, Nepal

- Developed FloraNepal (a.floranepal.com), a nursery management web application, as the primary project deliverable using Laravel, Blade, Tailwind CSS, and Alpine.js
- Implemented token-based authentication with Laravel Sanctum and Google SSO via Laravel Socialite, enabling secure multi-provider user login
- Engineered an SPA-like navigation pattern using Alpine.js and custom **X-Dashboard-Navigate** headers, reducing full-page reloads across the dashboard

PROJECTS

Traffic Hotspot Detection System 2026
[!\[\]\(19d44b37fb4fa155bf9d60c77a3d3cb2_img.jpg\) GitHub](#)

- Implemented DBSCAN from scratch using Haversine distance to detect 43 accident hotspots across Leeds (1,498 records), with epsilon selected automatically via the k-distance elbow method rather than hardcoded
- Built an ID3 decision tree from scratch (no scikit-learn) for accident severity prediction, achieving 63.5% accuracy; identified class imbalance as the primary failure mode (14 Fatal records vs 1,062 Slight)
- Developed dual-language scrapers for English (Kathmandu Post) and Nepali (OnlineKhabar) accident articles using Devanagari keyword dictionaries for severity, time, and weather extraction
- Deployed full-stack Django application on Render with PostgreSQL, serving an interactive Leaflet.js hotspot map and Chart.js analytics dashboard

Image Classification with Convolutional Neural Networks

2026

- Implemented and compared CNN architectures from scratch (LeNet, ResNet, AlexNet) in PyTorch on the MNIST dataset, evaluating accuracy, precision, recall, and F1-score
- Applied transfer learning with pretrained VGG16 and EfficientNetB0 (Keras/TensorFlow) for multi-class image classification, including partial fine-tuning of top backbone layers to improve accuracy on a domain-specific dataset

Learning Management System

2026

[Live](#)

- Designed a private, instructor-led LMS (Laravel, Alpine.js, Tailwind CSS) as a solo client project, replacing traditional marketplace-style LMS infrastructure with a focused, invite-only learning environment
- Built a token-based enrollment system to replace payment infrastructure entirely, giving instructors direct control over student access
- Enforced role-based access control at the middleware level across instructor, student, and admin flows, keeping route protection declarative rather than scattered through controllers
- Integrated Google OAuth for SSO, reducing signup friction and eliminating password storage; structured the codebase around the repository pattern to decouple business logic from the data layer

DwellCasa – Hotel Management System

2026

[Live](#)

- Designed a full-stack hotel management platform (Laravel, Alpine.js, Tailwind CSS) using the repository pattern to decouple business logic from data access across a multi-location tenancy model
- Implemented a status-based availability engine handling the full reservation lifecycle — bookings, check-ins, check-outs, and automated no-show resolution — preventing double-booking across concurrent requests
- Built a dual-track inventory system: consumable supplies with qty-based restocking and usage logs; durable equipment tracked by serial number, room assignment, condition history, and write-off status
- Enforced role-based access control across property locations — super admins switch between sites; admins and staff are scoped to their assigned property only

AI-powered Recommendation System for E-commerce Store

2025

 [GitHub](#)

- Built a semantic retrieval engine using Sentence-BERT embeddings and cosine similarity — the same core retrieval mechanism used in RAG pipelines — to surface contextually relevant products
- Built full-stack e-commerce platform (PHP/Python, MySQL, HTML/CSS) with user authentication, inventory management, and item filtering
- Integrated a user rating system to capture feedback for future recommendation refinement